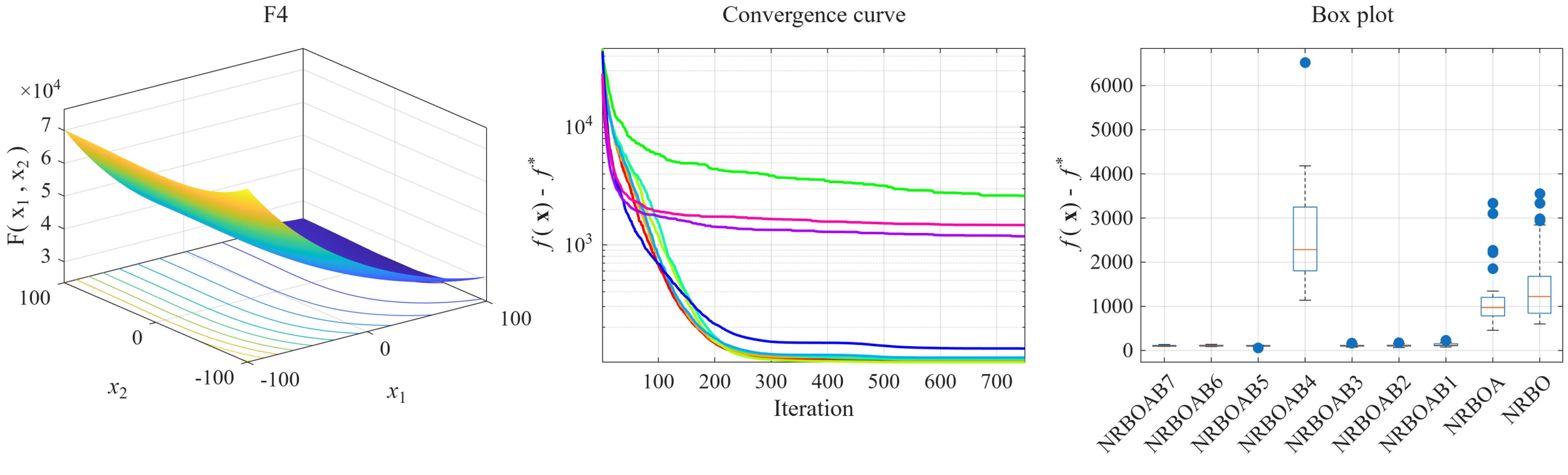

F4
×10^4
7
6
5
4
3
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^4
10^3
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
6000
5000
4000
3000
2000
1000
0
f(x) - f*
NRBOAB7
NRBOAB6
NRBOAB5
NRBOAB4
NRBOAB3
NRBOAB2
NRBOAB1
NRBOA
NRBO
NRBOAB7
NRBOAB6
NRBOAB5
NRBOAB4
NRBOAB3
NRBOAB2
NRBOAB1
NRBOA
NRBO